

Do you have time for a survey? Challenges and Lessons Learned from the Recruitment Process for an Online Survey

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ABSTRACT

Regardless of study design, recruitment is always an important challenge for researchers. In the current paper, we would like to share some lessons learned in the recruitment techniques for an online survey study. In order to recruit for a survey study on Esports, the authors decided to try and recruit from a diverse population. To this end, they recruited both from their personal networks and via social media. While they were unable to recruit the desired number of participants (30–50), they were able to learn a great deal about what worked and what did not in the recruitment process. Some of the issues encountered involved survey length and timing issues. Also, they learned that researchers need to be mindful about the limitations of the social media sites they use. We hope that sharing these lessons will be helpful to future researchers as they plan future research.

CCS CONCEPTS

• Human-centered computing → User studies.

KEYWORDS

Esports, survey methodology, study recruitment, online survey

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1 INTRODUCTION

Participant recruitment is an important challenge for any study, regardless of design. Interestingly, discussion of participant recruitment seems to rarely be published or discussed outside of classroom contexts. In the classroom, young researchers are taught the pros and cons of various types of recruitment. For example, a student sample from the university where the research is taking place is often cheap and accessible, but is not necessarily generalize-able

to larger populations. On the other side, recruitment from diverse adult populations seems ideal, but convincing people to participate without pay or with low pay is difficult and can be very expensive. But if pay becomes higher then are we, as researchers, manipulating participants into a situation where they feel as though they can not ask to withdraw? Such a scenario begins to cause ethical problems. All of these and more are important considerations during the research design process.

In the following paper we would like to illustrate what recruitment techniques were used for one study design and the lessons learned from that experience, so that other researchers can avoid our mistakes. The general goal was to assess the training procedures currently in use by Esports industry professionals. We also wished to assess what training areas are perceived to be the most and least important, and where gaps are perceived. The research project of interest was exploratory in nature, and as such it covered a variety of areas. This study was an online survey of about 15–30 questions (some of the questions were available depending on prior answers) which took the participant under 30 minutes to complete.

The following paper describes the design and data collection process, specifically focusing on how the researchers struggled to collect data. We also speculate the reasons for this difficulty.

2 RELATED WORK

The 30 question survey of the current study was based on a number of preexisting research papers [6–9, 13, 14, 16, 18]. The majority of these papers involved the use of online surveys in some fashion [7, 8, 13, 16, 18]. The remaining studies consisted of reviews and meta-analyses [6, 9, 14].

2.1 The Domain of Esports

In order to further communicate the aim of the study and its design, the following section summarizes past work related to the topic area.

Responses from one survey and interview paper found that most Esports participants believed that physical exercise helped their performance and reported the physical training took up 1 hour of their daily average of a little over five training hours per day [8]. However, another paper found that forty percent of collegiate Esports players did not participate in any physical exercise [7]. There was also a review which heavily cited the two previously mentioned papers [7–9]. They recommended further exploration into a concept called "Exergames," which are video games which are controlled via physical body movements [9]. In order to address this, the current survey included a question about physical training.

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In the current study we also asked multiple questions about training programs and Esports player's interest in them. This was inspired by Rudolf et al. [16], who looked at the intersection of Esports and health. Part of their study involved asking about training routines. Of their participants that said they had taken part in a tournament or league, more than half said they used structured training programs and around thirty percent said that they had a coach [16].

More empirical research has been conducted via a combination of survey and interviews. One paper focused on different game types/genres and what their informational needs were for replays or post-game visualizations. Generally, they found that different game genres had different desires and priorities when reviewing their game play [18]. As such, we also included questions asking about "cognitive or 'Brain game'" training in order to examine a research area that seems to be rarely discussed in the context of Esports [6, 14].

The strongest influence on the current study was Nagorsky and Wiemeyer's 2020 paper [13]. They asked participants to rate the importance of several skill dimensions on a Likert scale. We repeat this using their Integrative model, but include a forced-choice rating portion, in order to force participants to choose which item is the most/least important as they can rate things as equally important via Likert scales [13].

2.2 Online Surveys

Online surveys have become a research staple because of their speedy and efficient data collection of geographically diverse populations [4, 5, 10]. This is true of both quantitative and qualitative surveys [4]. In addition, social media has been found to be a solid tool for participant recruitment, both in the context of paid and unpaid surveys across multiple domains [1, 3, 10, 13, 16]. Also, much of this research was successful using websites like Facebook, LinkedIn, and Twitter, as well as emailing and reaching out to specific communities on these social media platforms [1, 3, 10, 13, 16].

After taking all of this previous research into account, we created an online survey consisting of 30 questions. In order to recruit for the survey, we decided to recruit via personal contacts and social media.

3 METHOD

Our aim was to recruit 30-50 participants from both the professional field of Esports as well as fans. The survey was intended to remain open from May 15 to June 15 2022 initially, but the authors planned to extend to the end of July if necessary. We based our timeline for data collection on prior research. For example, another project collected data for five weeks and was able to recruit over two hundred participants [10]. For a within domain comparison, another study on Esports collected data for three months and had a sample of over one thousand, although they had an incentive for participation (participants were entered into a drawing for a gift card) [16]. We expected to get a reasonable number of participants, even if the numbers would not have reached the heights of these prior studies, due to recruiting both a specific population and not offering a prize as incentive [10, 16].

Our primary research questions were as follows: Are professionals training by following a coach/program? What do these programs consist of? Are they interested in a program/coaching? How does the coaching/program differ from independent training? We also wanted to explore training that goes beyond simply playing the target game, for example physical exercise and cognitive training. In order to collect a broad perspective, we also collected data from Esports fans, as it may be interesting to compare their perceptions with the professionals. All of these questions were examined in a survey hosted on LimeSurvey.



Figure 1: The flyer created for the Esports survey

We planned to recruit participants using a variety of methods in order to get a diverse sample. Our recruitment techniques included social media recruitment, personal networks, and cold emailing. The following discusses more details of each of these areas.

One of our recruitment techniques was based on flyer distribution (see Figure 1). We intended to advertise at a local conference via the flyer. We also intended to hang flyers up in the university where the study was based. This flyer was also posted as an image to various social media websites. Both authors felt that they had some personal contacts which would be helpful for recruitment. We also asked several other researchers for their personal contacts. In addition to recruiting individuals, we also planned on recruiting and making connections with other Esports related entities via cold emailing. This included a number of Esports leagues, teams, and research groups. We also planned to post survey recruitment messages in multiple Discord communities.

Social media is widely used by people [2], and, as previously stated, has also been used for recruitment by researchers in other domains [1, 3, 10] and within Esports [13, 16]. We designed a "recruitment copy" to post on text-based sites like, Twitter, Reddit, LinkedIn, and Facebook. It included hashtags for popular Esports games and we tried to select the most popular, official hashtags from various Esports leagues. This text-based recruitment message

varied slightly in wording and hashtags used, in order to appeal to the platform they were posted on. Most of these text-based posts also included an image of the survey flyer (Figure 1) or linked to recruitment videos made by the authors. The following post was made on LinkedIn and included the survey flyer (Figure 1) :

"Are you involved in #Esports? As part of an ongoing project at HCI Lab @ MUNI we are recruiting for a research study on Esports Training procedures. We would like to hear from you, whether you're a fan, a player, an organizer, or a coach. Even if you aren't interested in Esports yourself, we would appreciate you sharing this message with any friends. Thank you so much! (there was an emoji here).

#Dota #PUBG #CSGO #Fortnite #R6Siege #LOL #Valorant #RocketLeague #Starcraft2 #HSEsports #FIFA22 #ApexLegends #MortalKombat #OWL2022 #StreetFighter #SSB #training #research #recruiting "

4 CHALLENGES AND LESSONS LEARNED

We encountered several challenges during the recruitment process. The data collection began on May 15th 2022, and eventually concluded August 15th 2022. The survey page was visited 85 times, but most of the people who started failed to finish. After August 15th, there were 15 complete quality responses.

Different posts went live on different platforms on different days, leading the authors to estimate where particular participants were recruited from. However, these estimations were not perfect as participants were not asked about where they saw the recruitment information.

4.1 In person recruitment via flyer

Despite being able to gain some participants via scanning the QR code on the flyer (Figure 1), none of those individuals fully completed the survey. The flyers were also left at a convention center during a game development conference. Because the situation was not controlled, it is unknown how many individuals saw the flyer.

It was not possible to administer flyers around the university. This is likely due to timing. The data collection was largely done over the summer, a time when many students, academics, and university staff are taking vacation and are not present at the university.

4.2 Recruiting through personal networks

In addition to recruiting participants via Esports communities, we also reached out to participants through personal networks. This was successful to some extent, as we estimate that many, if not most of our participants came from this area.

4.3 Social Media- Reddit, TikTok, Instagram, Twitter, LinkedIn, Facebook

We made many attempts to recruit via social media due to a belief that target participants would likely be on these platforms. Also, similar past studies have used similar tactics [13, 16]. The tactics we used differed slightly between platforms.

4.3.1 TikTok and Instagram. The three videos were created via TikTok and were posted to both TikTok and Instagram (we posted

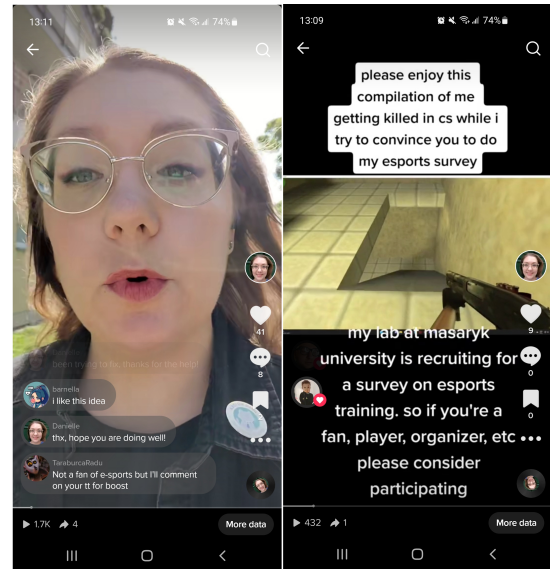


Figure 2: Screenshots two of TikTok videos used in recruitment

videos on Instagram instead of images because the app was emphasizing their videos [11, 17]) (see Figures 2 and 3). The TikTok videos reached a far wider audience, with the three videos reaching approximately 2500 views, 53 likes, 8 comments, and 2 saves. On Instagram the three videos reached just under 200 views and 15 likes.

One problem that arose when trying to recruit on Instagram and TikTok was the issue of exterior links. On Instagram, we were able to add links into the poster's profile (but not in the description of the video). However, on TikTok we were not able to link to the survey in the video description or on the user's profile (primarily due to a lack of followers as users must reach a certain threshold of followers to add external links). The only thing we were able to link on TikTok was an Instagram page. Therefore, we ended up asking participants on TikTok to click on the posting user's profile and then click through to their Instagram profile for the link.

4.3.2 Twitter, Reddit, LinkedIn, Facebook. On Twitter, we took advantage of multiple Twitter accounts. Both authors had their own account in addition to the official laboratory account. This meant that the posts were made and reposted across all three accounts. Further, the videos made for TikTok and Instagram were also linked here (see Figures 2 and 3). There were 10 total independent tweets made from the three accounts over the three months, and each of these tweets were retweeted by the accounts they did not originally come from. There were also approximately one follow-up tweet and/or quote tweet for each individual independent tweet.

On Reddit we made 4 posts on the subreddit r/samplesize (May 18, May 24, June 20, July 1). This subreddit is primarily focused on users both recruiting and participating in surveys. We also made three other attempts on other subreddits but were immediately deleted as many subreddits have rules against study recruitment. Researchers were also unable to get permission from moderators. Eventually

the researchers discontinued recruitment on all subreddits except for r/samplesize.

Much like Twitter, on LinkedIn both the authors and the lab accounts advertised the study, posting the recruitment information as well as the flyer and videos made for TikTok and Instagram (see Figure 3). There were six total independent posts made from the three accounts made over the three months, and each of these posts were reposted by the all accounts they did not originally come from. There was some engagement from users outside of the lab members, mostly from other academics. This is likely due to our personal circles primarily involving other academics.

On Facebook two posts were made, one included the survey flyer. These posts only attracted a single "Like". This is likely due to inactivity on the poster's Facebook page and a lack of "friends".

For all social media sites we attempted to add hashtags that would get the posts on other people's timelines, however that was unsuccessful. This was possibly due to a lack of following on the researcher's accounts or not selecting the right hashtags.

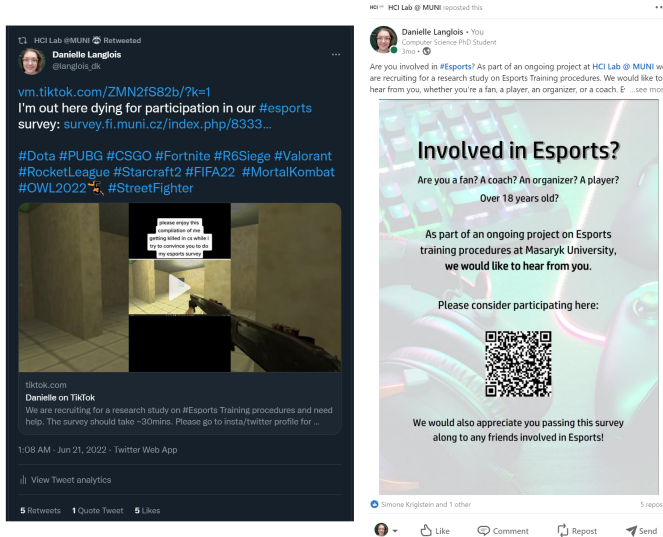


Figure 3: Screenshot of Twitter and LinkedIn posts using the flyer and TikTok videos for recruitment

4.4 Cold Emailing

We also directly emailed information about the survey to approximately fifty entities, including Researchers, University ESports teams, professional ESports teams, ESports leagues, ESports arenas, and other ESports groups. Although many of these emails reached people who were unavailable (due to the time of year, a full workload, etc), some of these emails reached individuals who were willing and able to participate or pass along the survey's information.

4.5 Discord

We posted in a few Discord communities that the authors were a part of. One was research related, one was oriented towards a game

development project, one was a personal Discord server, and the last was a podcast community. These posts yielded several participants, particularly from the research related Discord.

5 REFLECTION AND RECOMMENDATIONS

Despite the fact that we only got 15 participants when we wanted 30-50, we still got some interesting results. For example, It seemed that one of the most important overall categories to our participants was teamwork. This seems reflective of prior findings [12]. Further, most participants that were specialists we surveyed said that they had little to no formal training resources, and even said that they did not think such a thing existed. This is counter to prior research findings [16].

In the future, there are a number of things that we would do differently in recruitment for online surveys as a result of this experience:

- Be more strategic about data collection timing. Specifically, trying to collect data over the summertime months was a challenge, at least partially due to the academic calendar.
- Be thoughtful about the survey content itself. We noticed that many participants were mostly dropping off early on, many at the consent form, and did not even start the survey in earnest. We do not know what exactly caused participants to drop out here. Perhaps the consent form overwhelmed and confused participants, or perhaps it was the requirement that participants be over age 18. Also, the survey was likely too long for participants to be willing to do it for free. The issue here is that there is not really a way to know exactly what went wrong, but it is worth keeping in mind that good survey design is imperative.
- Track where participants are coming from, so that we could better estimate which platforms are worth spending recruitment time on.
- Try to provide an additional reward for participating in the study. While most previous research did not mention payment or rewards for participation, one study offered that participants would be entered into a drawing for a gift card [16]. Adding a dimension like this would make participation more appealing.
- Consider approaching the study itself differently, perhaps recruiting a "Survey Panel" via a paid service.
- Provide an easy way for people to reach the survey website. In this situation, it was sometimes difficult for participants to navigate to the survey link itself. This is likely a result of social media sites wanting to keep people on their platforms and actively discourage potential participants from leaving [15]. Asking participants to click onto your profile, then another social media profile to get to the survey link is a lot to ask. One plausible option to this end would be adding a QR code or an easy to type in shortcut to the survey.
- Consider social media reach realistically. It is likely harder to be seen on social media if you have a smaller following, and as such social media recruitment might not be as plausible for those who are not regularly producing content. Also, researchers planning on using social media for recruitment

should probably make sure to do plenty of background work on how to best optimize social media visibility [19].

6 CONCLUSION

Overall, we attempted to recruit 30-50 participants for this survey. We used a combination of methods to try and recruit people, primarily via personal networks and social media recruitment. These methods were partially successful, but we encountered a number of challenges. Some of these challenges included a survey that was too long, difficulty getting people onto the survey web page, and timing issues. This paper points to the issues we have encountered and suggest improvements for future participant recruitment using online surveys as data collection.

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